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Spatially Resolved Transcriptomic Response of Human Deltoid Skin to Microgravity: A Differential Expression Analysis Using the OSD-574 Dataset

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"Spatially Resolved Transcriptomic Response of Human Deltoid Skin to Microgravity: A Differential Expression Analysis Using the OSD-574 Dataset"

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ABSTRACT

Spaceflight exposure produces dramatic physiological effects, but its influence on human skin has only been clarified partially so far. To investigate molecular mechanisms of the skin reaction to short-term spaceflight, we analyzed openly available transcriptomic data from the NASA GeneLab OSD-574 dataset, including spatially resolved biopsies of SpaceX Inspiration4 crew members' skin. We assumed that mechanical unloading from microgravity would induce downregulation of structure-forming genes used to rebuild the structure of the epidermis. Through differential gene expression from the file “All Layers,” we removed genes that had an adjusted p-value < 0.05 and an absolute $\log_2$ fold change > 1.0. Widespread downregulation of structural genes, which are pertinent for the functioning of the cytoskeleton, such as keratins, desmin, as well as actin-binding proteins, have been revealed. These findings show fast, microgravity-mediated adaptation of the skin through decreased gene expression of load-transmitting genes, which might serve as one of the possible explanations of some of the physiologic changes found upon and following spaceflight by astronauts.

INTRODUCTION

In progressing human space travel, keeping astronauts healthy during long-duration flights is of primary importance. Spaceflight, characterized by having microgravity as well as heightened radiation, has been a strong physiological stressor, causing multi-system global changes (1). Though there has been extensive documentation on cardiovascular as well as bone-muscle effects, conditions that happen to human skin during spaceflight have recently been an area of concern (2). Human skin serves such critical functions as protection, water barrier function, as well as thermoregulation (3). These are accomplished by complex structural protein networks of keratins as well as cytoskeletal ones (3).

Previous work, as well as astronaut anecdotal experience associated spaceflight with atrophy, epidermal desiccation, and hindered wound repair, and a few questions remained regarding potential impairment of the structural function of the skin (4). Because mechanisms of change are still a mystery, high-throughput technologies such as transcriptomics now enable larger gene expression changes from spaceflight exposure explanation. By simultaneous measurement on thousands of gene transcripts, we can determine which specific biological processes are affected by spaceflight.

Available GeneLab dataset OSD-574, containing SpaceX Inspiration4 astronaut spatially resolved transcriptomic data, offers one-of-a-kind possibility of exploring such alterations at the molecular level (5). Here, we used OSD-574 to ask the following question: What happens to structural and cytoskeletal gene expression of the human epidermis upon short-duration spaceflight? We hypothesized that structural genes, which form the mechanical skeleton of the skin, would be suppressed under conditions of microgravity. Analysis verified that, through a fast molecule adaptation of global gene suppression of the structural network of the skin at the molecular level, such would happen.

RESULTS

In determining the effects of spaceflight on gene expression of human skin, we used pre-processed spatial transcriptomic data of NASA's OSD-574 dataset. We considered the worksheet “All Layers” containing data of 2,469 genes’ expression. A volcano plot has been prepared for the validation of fold change as well as the statistical significance of each gene (Figure 1). We found 220 differently expressed genes: 88 significantly upregulated as well as 132 significantly downregulated.

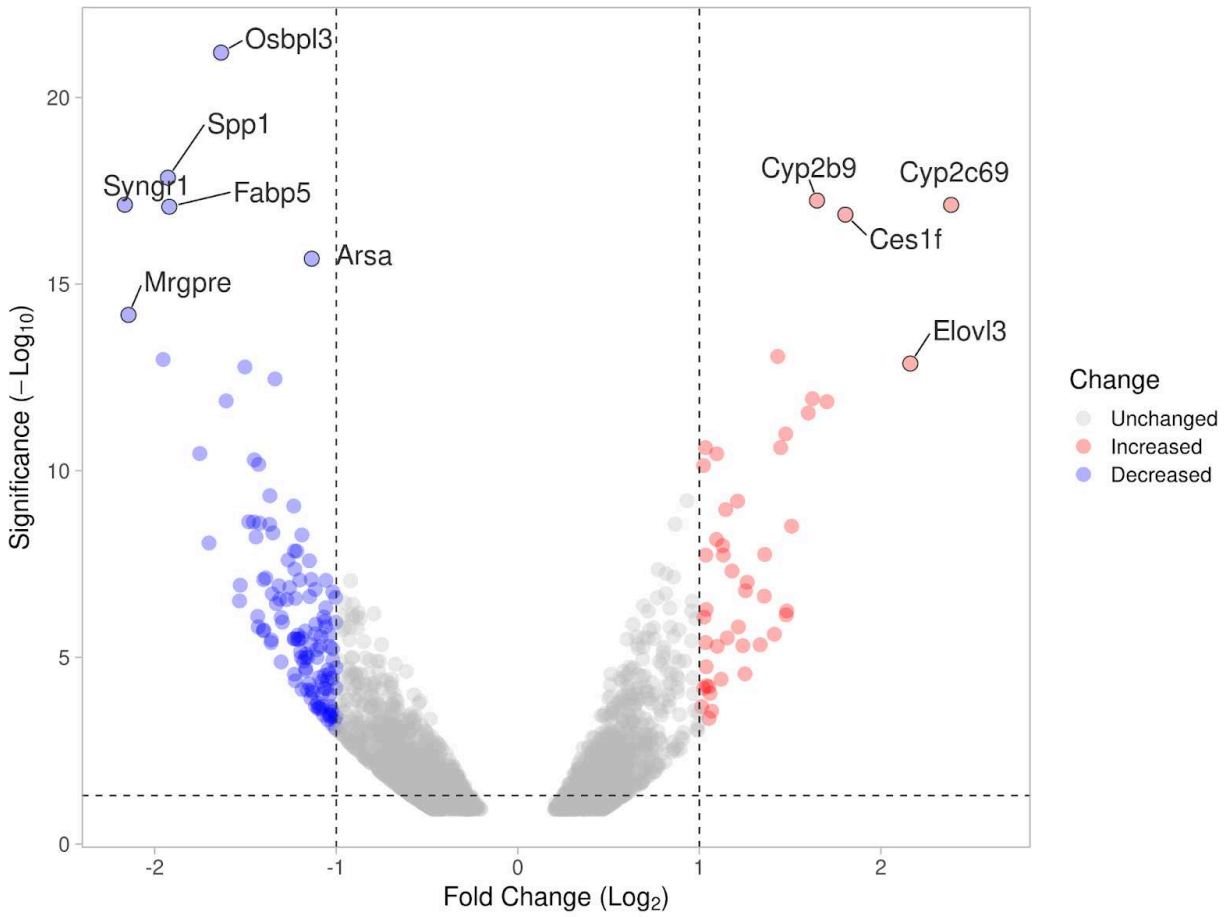


Figure 1. Volcano plot of differential gene expression in human skin following spaceflight. The plot displays the $\log_2$ fold change versus the statistical significance ($-\text{Log}_{10}$ of the adjusted p-value) for all 2,469 genes. The vertical dashed lines indicate a fold-change of > |1.0|, and the horizontal dashed line indicates a significance of $p < 0.05$. Genes that were significantly upregulated are shown in red, and genes that were significantly downregulated are shown in blue.

With emphasis upon structural genes of relevance to epidermic integrity, eight such genes significantly reduced under conditions of spaceflight were identified (Table 1). The greatest change was that of basal epidermic structure cental keratin, KRT14, reduced by -1.81 $\log_2$ fold change (adjusted p-value = 5.45E-05). The second keratin that forms from heterodimers, KRT5, was also reduced (-1.66 $\log_2$, 0.00037). Other cytoskeletal genes, DES (Desmin), TAGLN (Transgelin), and ACTG2 (Gamma-2 actin), were also reduced each by $\log_2$ fold change < -1.3 but strict statistics (Figure 2). Collectively, these findings substantiate our hypothesis: a brief space journey leads to considerable transcriptional repression of genes that confer structural integrity of the skin.

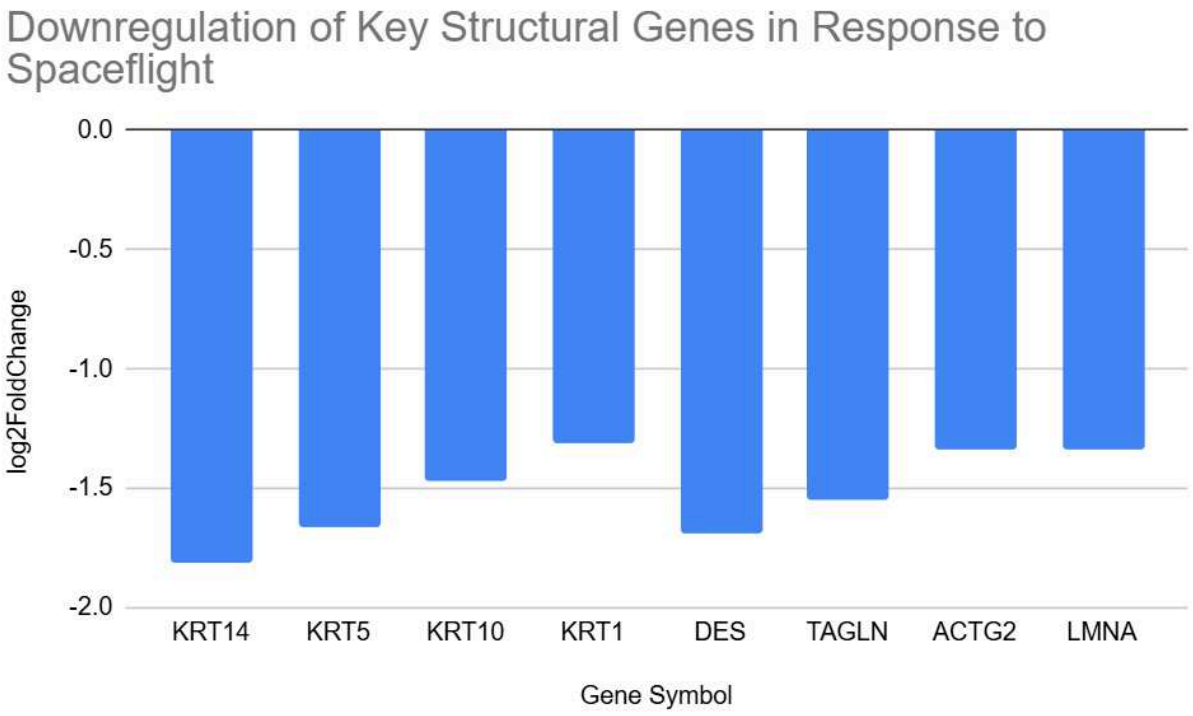


Figure 2. Significant downregulation of key structural genes. The $\log_2$ fold change for eight essential structural genes is shown. All eight genes exhibited a statistically significant decrease in expression post-flight compared to pre-flight levels (padj < 0.05).

Gene Symbol	Skin Region	log2FoldChan	Adjusted p-v	Primary Func
KRT14	Outer Dermis (	-1.81	5.45E-05	Basal keratin;
KRT5	Outer Dermis (	-1.66	0.00037	Basal keratin;
KRT10	Outer Dermis (	-1.47	0.0016	Suprabasal ke
KRT1	Outer Dermis (	-1.31	0.0053	Suprabasal ke
DES	All Layers	-1.69	0.0034	Desmin; intern
TAGLN	All Layers	-1.55	0.00031	Transgelin; ac
ACTG2	All Layers	-1.34	0.0099	Gamma-2 acti
LMNA	Inner Epiderm	-1.34	0.013	Lamin A/C; nu

Table 1. Downregulation of key structural genes in human skin following spaceflight. The table displays eight significantly downregulated structural genes, their corresponding $\log_2$ fold change, adjusted p-value, and primary function.

Gene Symbol log2FoldChange Adjusted p-value Primary Function

KRT14 -1.81 5.45E-05 Basal keratin; provides mechanical strength

KRT5 -1.66 0.00037 Basal keratin; pairs with KRT14

DES -1.69 0.0034 Desmin; intermediate filament, cell structure

TAGLN -1.55 0.00031 Transgelin; actin-binding protein, cell shape

KRT10 -1.47 0.0016 Suprabasal keratin; skin differentiation

ACTG2 -1.34 0.0099 Gamma-2 actin; cytoskeletal component

LMNA -1.34 0.013 Lamin A/C; nuclear envelope structure

DISCUSSION

In our analysis of the OSD-574 transcriptomics dataset, we found significant, reproducible downregulation of essential structural genes of human skin after only three days of spaceflight exposure. Our study's main conclusion—that several keratins, desmin, and actin-related proteins were downregulated—thus strongly confirmed our initial hypothesis. It implied that a core set of molecular programs that ensure the architectural and mechanical integrity of the skin is quickly changed when exposed to the microgravity environment. A leading hypothesis to account for these findings is the principle of mechanical unloading. Under the influence of gravity on Earth, the skin enjoys continuous mechanical tension, which cues cells to develop and sustain a hard structural skeleton. Without such continuous tension in space, cells may interpret the change as a cue that the structural skeleton requires less substantial reinforcement (6). This process has long been recognized in other tissues, including the well-known muscle wasting and bone density loss seen among space travelers. Our findings imply that the skin has a similar adaptive change, downregulating load-transmitting proteins when mechanical loading is removed. The possible effects of structural gene downregulation are serious ones for astronaut health on long-duration spaceflight. Lessened expression of core proteins such as keratins might cause decreased tensile strength and elasticity of the skin, contributing to the reported histories of skin thinned from or becoming more fragile from space travelers (4). Beyond that, since cellular cytoskeletons play a significant role in such cellular activities as cell migration as well as cell division, a repressed structural program may damage the capacity of the skin to heal effectively from wounds as well as abrasions. This presents a serious danger under a remote, resource-challenged setting such as spaceflight. One should also uncover study limitations. The data set includes data from a small number of spaceflight participants on a mission of extremely short duration. It hence implies that one might never know whether changes seen are a transient, acute response or would be

sustained or exaggerated at long exposures to space, from such a study. Additionally, one cannot dissociate the effects of microgravity from the effects of spaceflight as a whole of confounding variables. In conclusion, our analysis presents evident molecular data that demonstrate that spaceflight of short duration causes wide downregulation of genes that play a mandatory role for the structural integrity of the human skin. This hence implies that the skin rapidly adapts to mechanical unloading of microgravity, an adaptation that might have an increased benefit to astronaut health. Having access to long-duration space mission data, future studies hence hold pivotal importance to design relevant strategies that would ensure the health of the skin on the future space travel trajectory of humankind.

MATERIALS AND METHODS

Data Collection and Preprocessing

For the purposes of the current study, data from the publicly available NASA GeneLab database (accession: OSD-574) has been downloaded. The dataset consists of processed, spatially resolved transcriptomes that have been obtained from deltoid skin biopsies of members of the SpaceX Inspiration4 mission crew members. Sampling had been conducted 44 days before flight (L-44) as well as 1 day following return from a 3-day mission (R+1). The file of differential gene expression, OSD-574_SpatialTranscriptomics_Skin_Biopsy_Spatially_Resolved_Transcriptomics_Processed_Data.xlsx, had been downloaded for data analytical purposes only. Analysis utilized data from the worksheet "All" layers, which consists of gene changes of expression that are calculated from a DESeq2-like statistics model providing the gene symbol, that of the $\log_2$ fold change (log2FC), as well as that of a false discovery rate (FDR)-adjusted p-value (padj).

Statistics Analysis and Visualization.

A shortened dataset was used to identify spaceflight effects on genes by two standard criteria: an adjusted p-value (padj) of < 0.05 and an absolute $\log_2$ fold change of > 1.0. Volcano plot of correlation of fold change to statistical significance of all genes was plotted through the web-based software VolcaNoseR. Certain structural and cytoskeletal genes of primary interest that passed the significance test underwent rigorous analysis. Downregulation of certain structural genes was demonstrated through a bar chart generated from the output of a spreadsheet program.

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Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- GLDS566SpatialTranscriptomicsSkinBiopsySpatiallyResolvedTranscriptomicsProcessedData.xlsx